

DAVID ANDERLE

Computer Science Student | Quantitative Finance | Python, Data Analysis & Research Software
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PROFILE & RESEARCH INTERESTS

Informatics student building toward quantitative finance through Python, financial modeling, data analysis and reproducible research software. Research interests include market risk, financial data quality, leverage and liquidity mechanisms, and software that makes quantitative claims inspectable.

EDUCATION

Czech University of Life Sciences Prague - Bachelor's Degree in Informatics (expected Jun 2027) Sep 2024 - Present
Prague, Czechia. Faculty of Economics and Management. Relevant foundation: Programming, Algorithm Development, Computer Architecture, Mathematical Logic and Graph Theory, Mathematics.

University of Wisconsin-Madison - Non-Degree Exchange Program - Finance, Economics and Quantitative Methods Sep 2025 - May 2026

Madison, Wisconsin, USA. Selected completed coursework: Business Valuation, Derivative Securities, Financial Modeling in Excel, Simulation & Probabilistic Modeling.

VOŠ a SPŠ Elektrotechnická Františka Křížíka - Secondary Technical Diploma in Information Technology Sep 2020 - May 2024

Prague, Czechia. Four-year technical program covering programming, operating systems, networking, databases, computer hardware and applied automation/electronics laboratories.

RESEARCH & TECHNICAL PROJECTS

Volatility Cascade Engine - Independent Python research software project Mar - Jul 2026

davidanderle.com/work/volatility-cascade-engine/

- Built a deterministic synthetic-market simulation of leverage, margin thresholds, forced liquidation, overlapping portfolios and endogenous price impact across six funds and five assets.
- Designed and published a 15-scenario shock sweep, deterministic seed, model specification, scenario data, unit tests, validation checks, figures and a versioned technical report.
- Preserved negative results and explicit boundaries: the prototype is synthetic and uncalibrated, does not predict crashes, and does not reproduce a complete broker margin system.

Merkle Commitments with Poseidon in Rust - Collaborative Rust/arkworks prototype Apr 2026 - Present

github.com/davidanderle1/merkle-poseidon

- Updated Poseidon hashing, Merkle-tree, membership-proof and README components in a public collaborative repository while preserving the collaborator-authored core implementation boundary.
- Documented that the current Poseidon parameters are demonstration-only and that the threshold comparison is computed natively rather than enforced in zero knowledge.

SELECTED PUBLICATION

Anderle, David. "Protecting Retail Investors: A Framework for Transparency and Risk Controls in High-Volatility Trading." Proceedings of the Tenth Annual Wisconsin Ideas Conference, University of Wisconsin-Madison, Apr 2026, p. 109. *Policy memo examining information asymmetry, transparency and risk controls in high-volatility retail-trading environments. No peer-review claim is made.*

PROFESSIONAL EXPERIENCE

Hewlett Packard Enterprise - IT Training Placements May 2022 & May 2023

Prague, Czechia

- Completed two short school placements covering enterprise infrastructure, server hardware, basic systems administration, HPE GreenLake and hybrid-cloud concepts.

LEADERSHIP & SERVICE

Associated Students of Madison - Student Council Representative Feb - May 2026

Madison, Wisconsin, USA

- Represented student interests and reviewed proposals involving university policy, resource allocation and student services.

CULS Racing - Sponsorship & Business Strategy Team Member Oct 2024 - Jul 2025

Prague, Czechia

- Supported sponsorship outreach, partner-facing materials, business-plan development and the Formula Student business presentation.

TECHNICAL SKILLS & LANGUAGES

Programming / Data: Python, SQL, Rust, Git, PostgreSQL, DuckDB, APIs, Excel

Quantitative / Finance: Data Analysis, Statistics, Financial Modeling, Valuation, Risk Analysis, Scenario Analysis

Languages: Czech (native), English (full professional), German (elementary)

Research Practice: Reproducible workflows, explicit assumptions and limitations, validation checks, versioned artifacts, technical writing